PARAMETER EXPANSION

Indirection

Parameter	What does it do?
`${!VAR}`	Access a variable based on the value of `VAR`.
`${!VAR*}`	Expand to `IFS` separated list of variable names starting with `VAR`.
`${!VAR@}`	Expand to `IFS` separated list of variable names starting with `VAR`. If double-quoted, each variable name expands to a separate word.

Replacement

Parameter	What does it do?
`${VAR#PATTERN}`	Remove shortest match of pattern from start of string.
`${VAR##PATTERN}`	Remove longest match of pattern from start of string.
`${VAR%PATTERN}`	Remove shortest match of pattern from end of string.
`${VAR%%PATTERN}`	Remove longest match of pattern from end of string.
`${VAR/PATTERN/REPLACE}`	Replace first match with string.
`${VAR//PATTERN/REPLACE}`	Replace all matches with string.
`${VAR/PATTERN}`	Remove first match.
`${VAR//PATTERN}`	Remove all matches.

Length

Parameter	What does it do?
`${#VAR}`	Length of var in characters.
`${#ARR[@]}`	Length of array in elements.

Expansion

Parameter	What does it do?
`${VAR:OFFSET}`	Remove first `N` chars from variable.
`${VAR:OFFSET:LENGTH}`	Get substring from `N` character to `N` character. (`${VAR:10:10}`: Get sub-string from char `10` to char `20`)
`${VAR:: OFFSET}`	Get first `N` chars from variable.
`${VAR:: -OFFSET}`	Remove last `N` chars from variable.
`${VAR: -OFFSET}`	Get last `N` chars from variable.
`${VAR:OFFSET:-OFFSET}`	Cut first `N` chars and last `N` chars.

Case Modification

Parameter	What does it do?	CAVEAT
`${VAR^}`	Uppercase first character.	bash 4+
`${VAR^^}`	Uppercase all characters.	bash 4+
`${VAR,}`	Lowercase first character.	bash 4+
`${VAR,,}`	Lowercase all characters.	bash 4+
`${VAR~}`	Reverse case of first character.	bash 4+
`${VAR~~}`	Reverse case of all characters.	bash 4+

Default Value

Parameter	What does it do?
`${VAR:-STRING}`	If `VAR` is empty or unset, use `STRING` as its value.
`${VAR-STRING}`	If `VAR` is unset, use `STRING` as its value.
`${VAR:=STRING}`	If `VAR` is empty or unset, set the value of `VAR` to `STRING`.
`${VAR=STRING}`	If `VAR` is unset, set the value of `VAR` to `STRING`.
`${VAR:+STRING}`	If `VAR` is not empty, use `STRING` as its value.
`${VAR+STRING}`	If `VAR` is set, use `STRING` as its value.
`${VAR:?STRING}`	Display an error if empty or unset.
`${VAR?STRING}`	Display an error if unset.